H2 Math Summary

Example Questions

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1 Systems of Linear Equations

1.1 Three different blends of coffee beans – Blend A, Blend B and Blend C – are sold at both M Café and I Café. The usual selling prices of the blends are the same at both cafés. The total cost of buying one package each of Blend A, Blend B and Blend C is \$176. During a holiday sale the two cafés offered the following discounts:

Café	**Discounts given for each package**			**Total price after the discount**
	Blend A	**Blend B**	**Blend C**	
M Café	15%	10%	5%	\$161.80
I Café	7%	5%	2%	\$169.56

(i) Find the usual selling price of each package of Blend A, Blend B and Blend C coffee beans respectively. [3]

[MI 2025 Prelim P1 Q4]

(i) **[3]**

Let the usual prices be \$$a$, \$$b$, \$$c$.

$$a+b+c=176 \quad ---(1)$$

$$0.85a+0.90b+0.95c=161.80 \quad ---(2)$$

$$0.93a+0.95b+0.98c=169.56 \quad ---(3)$$

Using GC, $a=32,\ b=44,\ c=100$

$\therefore$ Blend A: \$32, Blend B: \$44, Blend C: \$100

1.2 A function f is defined by $\mathrm{f}(x)=ax^3+bx^2+cx+d$ where a, b, c and d are constants. The graph of $y=\mathrm{f}(x)$ has a turning point at $x=-1$ and passes through the points $(2, 10)$ and $(-2, -2)$. Given also that $\int_{-2}^{2} \mathrm{f}(x)\,\mathrm{d}x=16$, find the values of a, b, c and d. [4]

[SAJC 2025 Prelim P1 Q1]

Solution

$$\mathrm{f}'(x)=3ax^2+2bx+c$$

$$\mathrm{f}'(-1)=0$$

$$3a-2b+c=0 \quad ---(1)$$

$$\mathrm{f}(2)=10$$

$$8a+4b+2c+d=10 \quad ---(2)$$

$$\mathrm{f}(-2)=-2$$

$$-8a+4b-2c+d=-2 \quad ---(3)$$

$$\int_{-2}^{2} (ax^3+bx^2+cx+d)\,\mathrm{d}x=16$$

$$\left[\tfrac{a}{4}x^4+\tfrac{b}{3}x^3+\tfrac{c}{2}x^2+dx\right]_{-2}^{2}=16$$

$$(4a+\tfrac{8}{3}b+2c+2d)-(4a-\tfrac{8}{3}b+2c-2d)=16$$

$$\tfrac{16}{3}b+4d=16 \quad ---(4)$$

Using GC, $a=3,\ b=0,\ c=-9,\ d=4$

Two linear equations are independent if and only if neither equation is a linear combination of the other.

1.3 A sequence of real numbers, x_n, satisfies the recurrence relation

$$x_{n+1} = a(-3)^n + bn^2 + cx_n, \text{ for } n, a, b, c \in \mathbb{Z} \text{ and } n \geqslant 1.$$

Given that $x_1 = 1$, $x_2 = -8$, $x_3 = 70$, and $x_4 = -377$, find the values of a, b and c. [3]

Hence, find the values of x_{10} and x_{11}. [2]

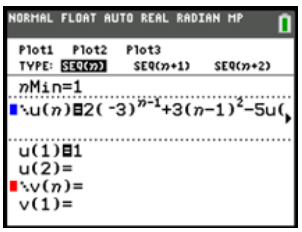
[HCI 2025 Prelim P1 Q2]

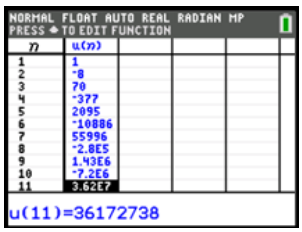
Solution

$$n = 1$$
$$x_2 = a(-3)^1 + b(1)^2 + cx_1$$
$$-8 = -3a + b + c \quad ---(1)$$
$$n = 2$$
$$x_3 = a(-3)^2 + b(2)^2 + cx_2$$
$$70 = 9a + 4b - 8c \quad ---(2)$$
$$n = 3$$
$$x_4 = a(-3)^3 + b(3)^2 + cx_3$$
$$-377 = -27a + 9b + 70c \quad ---(3)$$

Using GC, $a = 2,\ b = 3,\ c = -5$

$$x_{n+1} = 2(-3)^n + 3n^2 - 5x_n$$





Using GC,

$$x_{10} = -7\,210\,868$$
$$x_{11} = 36\,172\,738$$

1.4 The lines l_1 and l_2 have equations

$$l_1: \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix},$$

$$l_2: \mathbf{r} = \begin{pmatrix} 0 \\ 5 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix},$$

where λ and μ are parameters.

(i) Show that l_1 and l_2 are skew lines. [2]

[SAJC 2025 Prelim P1 Q3]

(i) **[2]**

$$1 - 2\lambda = 2\mu \quad ---(1)$$

$$2 + \lambda = 5 - 2\mu \quad ---(2)$$

$$2\lambda = -8 + 3\mu \quad ---(3)$$

Solve (1) and (2): $\lambda = -2,\ \mu = \frac{5}{2}$

Sub into (3):

$\text{LHS} = 2(-2) = -4$

$\text{RHS} = -8 + 3\left(\frac{5}{2}\right) = -\frac{1}{2}$

$\text{LHS} \neq \text{RHS},$ no solution

$\therefore l_1$ and l_2 do not intersect

$\because \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \neq k \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$ for any $k \in \mathbb{R}$

$\therefore l_1$ and l_2 are not parallel

$\therefore l_1$ and l_2 are skew (shown)

1.5 A movie theatre supports a charity event by providing discounted movie tickets at the following prices based on age group.

Age group	Discounted ticket price
$\leqslant 12$ years	\$6
13 to 49 years	\$9
$\geqslant 50$ years (senior citizen)	\$4

The event organiser plans to spend \$700 to sponsor 100 participants of which at least 35 of them are senior citizens. Find the possible number of participants for each age group. [4]

[TJC 2022 Prelim P2 Q1]

Solution

Let x, y, z be the number of participants aged $\leqslant 12$, 13 to 49 and $\geqslant 50$ years respectively.

$$\left.\begin{aligned} x+y+z=100 \quad &---(1) \\ 6x+9y+4z=700 \quad &---(2) \end{aligned}\right\} \Rightarrow \text{Using GC, } x=\tfrac{200}{3}-\tfrac{5}{3}z,\ y=\tfrac{100}{3}+\tfrac{2}{3}z$$

$$z \geqslant 35$$

$$\because x \geqslant 0$$

$$\tfrac{200}{3}-\tfrac{5}{3}z \geqslant 0$$

$$z \leqslant 40$$

$$\therefore 35 \leqslant z \leqslant 40$$

x, y, z are non-negative integers. Using GC table:

$$z=37 \Rightarrow x=5,\ y=58$$

$$z=40 \Rightarrow x=0,\ y=60$$

$$\therefore (x,y,z)=(5,58,37) \text{ or } (0,60,40)$$

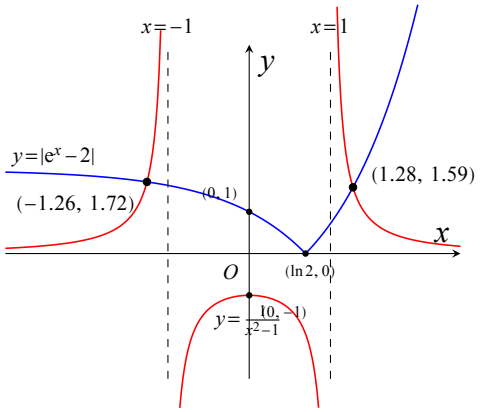
2 Inequalities

2.1 **(a)** Solve the inequality $|e^x - 2| > \dfrac{1}{x^2 - 1}$ using a graphical approach. [3]

(b) Without the use of a graphing calculator, solve the inequality $\dfrac{x+2}{x-1} \geqslant \dfrac{2}{x-3}$. [3]

[RVHS 2026 Prelim P1 Q2]

(a) **[3]**



Using GC, intersections at $x = -1.2581$, $x = 1.2770$

$\therefore x < -1.26$ or $-1 < x < 1$ or $x > 1.28$

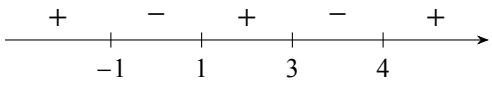
(b) **[3]**

$$\frac{x+2}{x-1} - \frac{2}{x-3} \geqslant 0$$

$$\frac{(x+2)(x-3) - 2(x-1)}{(x-1)(x-3)} \geqslant 0$$

$$\frac{x^2 - 3x - 4}{(x-1)(x-3)} \geqslant 0$$

$$\frac{(x-4)(x+1)}{(x-1)(x-3)} \geqslant 0$$



$\therefore x \leqslant -1$ or $1 < x < 3$ or $x \geqslant 4$

2.2 Use an algebraic method to solve the inequalities

$$\frac{x(5x+9)}{x^2+2x+5} > x \geqslant 3x-6.$$

[4]

[ACJC 2026 Prelim P1 Q1]

Solution

$$x \geqslant 3x-6$$

$$x \leqslant 3 \qquad ---(1)$$

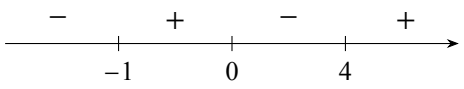
$x^2+2x+5=(x+1)^2+4>0$ for all real x

$$\frac{x(5x+9)}{x^2+2x+5} > x$$

$$5x^2+9x > x(x^2+2x+5) \quad (\because x^2+2x+5>0)$$

$$x^3-3x^2-4x<0$$

$$x(x-4)(x+1)<0$$



$$x<-1 \text{ or } 0<x<4 \quad ---(2)$$

$(1)\cap(2)$:

$\therefore x<-1$ or $0<x\leqslant 3$

2.3 **(i)** Solve the inequality $3\,|x^2-4| \leqslant |x+2|$. [4]

(ii) Hence, solve $\dfrac{3}{x^2}\,|1-4x^2| \leqslant \left|\dfrac{1+2x}{x}\right|$. [2]

[SAJC 2025 Prelim P2 Q1]

(i) **[4]**

$$3|x-2||x+2| \leqslant |x+2|$$

$x=-2$: $0 \leqslant 0$ (true)

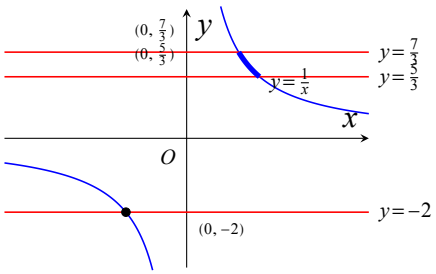
$x \neq -2$: $|x+2|>0$

$$3|x-2| \leqslant 1$$
$$-\tfrac{1}{3} \leqslant x-2 \leqslant \tfrac{1}{3}$$
$$\tfrac{5}{3} \leqslant x \leqslant \tfrac{7}{3}$$

$\therefore x=-2$ or $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$

(ii) **[2]**

$$\frac{3}{x^2}\,|1-4x^2| = 3\left|\frac{1}{x^2}-4\right| \quad (\because x^2>0)$$
$$\left|\frac{1+2x}{x}\right| = \left|\frac{1}{x}+2\right|$$
$$3\left|\left(\frac{1}{x}\right)^2-4\right| \leqslant \left|\frac{1}{x}+2\right| \quad (x \to \tfrac{1}{x})$$
$$\frac{1}{x}=-2 \text{ or } \frac{5}{3} \leqslant \frac{1}{x} \leqslant \frac{7}{3}$$



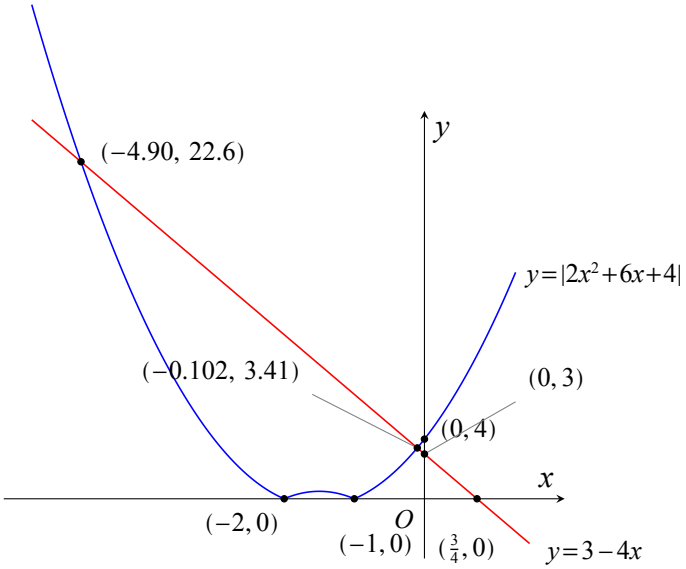
$\therefore x=-\frac{1}{2}$ or $\frac{3}{7} \leqslant x \leqslant \frac{3}{5}$

2.4 **(i)** On the same axes, sketch the curves with equations $y=|2x^2+6x+4|$ and $y=3-4x$, indicating any intercepts with the axes and points of intersection. Hence solve the inequality $3-4x<|2x^2+6x+4|$. [4]

(ii) Find the exact area bounded by the graphs of $y=3-4x$, $y=|2x^2+6x+4|$, $x=-3$ and $x=-1$. [3]

[SAJC 2019 Prelim P1 Q2]

(i) **[4]**



$$3-4x=2x^2+6x+4$$

$$2x^2+10x+1=0$$

$$x=\frac{-5\pm\sqrt{23}}{2}$$

$$\therefore x<\frac{-5-\sqrt{23}}{2} \text{ or } x>\frac{-5+\sqrt{23}}{2}$$

(ii) **[3]**

$$\begin{aligned}\text{Area}&=\int_{-3}^{-1}(3-4x-|2x^2+6x+4|)\,dx\\&=\int_{-3}^{-2}(3-4x-2x^2-6x-4)\,dx+\int_{-2}^{-1}(3-4x+2x^2+6x+4)\,dx\\&=\int_{-3}^{-2}(-2x^2-10x-1)\,dx+\int_{-2}^{-1}(2x^2+2x+7)\,dx\\&=\left[-\tfrac{2}{3}x^3-5x^2-x\right]_{-3}^{-2}+\left[\tfrac{2}{3}x^3+x^2+7x\right]_{-2}^{-1}\\&=\left(\tfrac{16}{3}-20+2\right)-(18-45+3)+\left(-\tfrac{2}{3}+1-7\right)-\left(-\tfrac{16}{3}+4-14\right)\\&=20\end{aligned}$$

3 Graphing Techniques

3.1 A curve C has equation $y=\dfrac{x^2-4x+1}{x+1}$, $x\neq -1$.

(a) Prove, using an algebraic method, that $y=\dfrac{x^2-4x+1}{x+1}$ cannot lie between two certain values, to be determined in exact form. [3]

(b) Sketch C, stating the equations of any asymptotes, and the coordinates of any turning points and any axial intercepts. [4]

[VJC 2025 Promo Q9(a), (b)]

(a) **[3]**

$$y=\frac{x^2-4x+1}{x+1}$$

$$yx+y=x^2-4x+1$$

$$x^2+(-y-4)x+(1-y)=0$$

No real solution for x: $(-y-4)^2-4(1-y)<0$

$$y^2+8y+16-4+4y<0$$

$$y^2+12y+12<0$$

$$(y+6)^2-24<0$$

$$\left(y+6-\sqrt{24}\right)\left(y+6+\sqrt{24}\right)<0$$

$$-6-2\sqrt{6}<y<-6+2\sqrt{6}$$

$\therefore$ y cannot lie between $-6-2\sqrt{6}$ and $-6+2\sqrt{6}$

(b) **[4]**

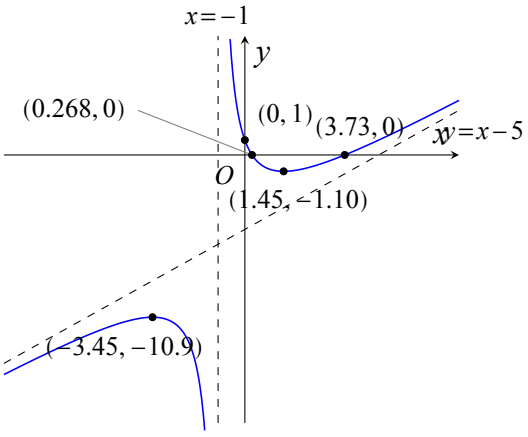
$$y=x-5+\frac{6}{x+1}$$

Asymptotes: $x=-1,\ y=x-5$

$x=0$: $y=1$

$y=0$: $x^2-4x+1=0 \Rightarrow x=0.268,\ 3.73$

Turning points: $(1.45, -1.10),\ (-3.45, -10.9)$



3.2 A curve C has equation $y=ax+b+\dfrac{2b-a}{x+3}$, where a and b are constants such that $a>0$, $b\neq\dfrac{1}{2}a$ and $x\neq -3$.

(a) If C has no stationary points, use differentiation to find the relationship between a and b. [3]

(b) It is now given that $b=a$.

(i) Find the x-coordinates of the turning points. [2]

(ii) Hence sketch C, stating the equations of any asymptotes and the coordinates of the axial intercepts and turning points. [3]

(iii) Verify that the point of intersection of the two asymptotes of C lies on the line $y=kx+3k-2a$.

Hence, using the graph in part (b)(ii), find the range of values of k in terms of a, such that the equation $x(k-a)+3k-2a-b-\dfrac{2b-a}{x+3}=0$ has real roots. [2]

[SAJC 2025 Promo Q7]

(a) **[3]**

$$\frac{dy}{dx}=a-\frac{2b-a}{(x+3)^2}$$

$$\frac{dy}{dx}=0: \quad a-\frac{2b-a}{(x+3)^2}=0$$

$$(x+3)^2=\frac{2b-a}{a} \qquad ---(1)$$

No stationary points $\Rightarrow \dfrac{2b-a}{a}<0$

Since $a>0$, $2b-a<0$

$\therefore a>2b$

(b)(i) **[2]**

$$b=a: \quad (x+3)^2=\frac{2a-a}{a}=1$$

$$x+3=1 \text{ or } -1$$

$$x=-2 \text{ or } -4$$

(b)(ii) **[3]**

$$y = ax + a + \frac{a}{x+3}$$

Asymptotes: $x = -3,\ y = ax + a$

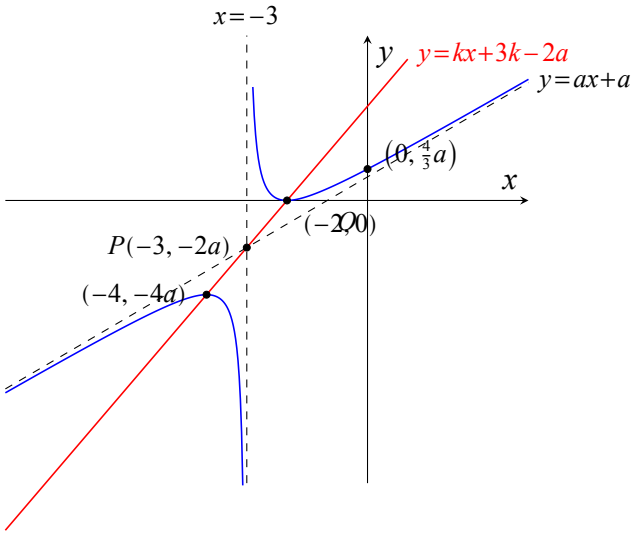
$x = -2 : y = 0; \quad x = -4 : y = -4a$

Turning points: $(-2, 0),\ (-4, -4a)$

$$x = 0 : y = \frac{4}{3}a$$

$$y = 0 : ax + a + \frac{a}{x+3} = 0 \Rightarrow (x+1)(x+3) + 1 = 0$$

$$x^2 + 4x + 4 = 0 \Rightarrow (x+2)^2 = 0 \Rightarrow x = -2$$



(b)(iii) **[2]**

$x = -3$ and $y = ax + a : \quad y = -3a + a = -2a$

Intersection of asymptotes: $P(-3, -2a)$

$x = -3 : \quad \text{RHS} = k(-3) + 3k - 2a = -2a = \text{LHS}$

$\therefore P$ lies on the line $y = kx + 3k - 2a$

$$x(k-a) + 3k - 2a - b - \frac{2b-a}{x+3} = 0$$

$$kx + 3k - 2a = ax + b + \frac{2b-a}{x+3}$$

Line through P meets $C \Rightarrow k > a$

3.3 A curve C has equation $\mathrm{f}(x)=\dfrac{3x^2+6x+k}{x+2}$, where $x>-2$ and k is a non-zero real constant.
It is given that the function f is strictly increasing.

(a) Find the range of values of k. [3]

(b) Sketch the curve C, giving the coordinates of any points where C crosses the x- and y-axes and the equations of any asymptotes. [3]

(c) By adding a suitable curve to the graph of the curve C in (b), deduce the number of distinct real root(s) of the equation $(3x^2+6x+k)^2=(x^2+1)\,(9x^2+36x+36)$. [3]

[DHS 2025 Promo Q8]

(a) **[3]**

$$\mathrm{f}(x)=\frac{3x^2+6x+k}{x+2}=3x+\frac{k}{x+2}$$

$$\mathrm{f}'(x)=3-\frac{k}{(x+2)^2}=\frac{3x^2+12x+12-k}{(x+2)^2}$$

f strictly increasing $\Rightarrow$ $\mathrm{f}'(x)>0$

$3x^2+12x+12-k>0$ $\quad\because (x+2)^2>0$ for $x>-2$

$3(x+2)^2-k>0$

Since $3(x+2)^2-k>-k$ for all $x>-2$, $\quad -k\geqslant 0$

$k\neq 0 \Rightarrow k<0$

(b) **[3]**

$$y=3x+\frac{k}{x+2}$$

Asymptotes: $y=3x,\ x=-2$

$$x=0:\ y=\frac{k}{2}$$

$$y=0:\ 3x^2+6x+k=0 \Rightarrow x=\frac{-6\pm\sqrt{36-12k}}{6}=-1\pm\frac{\sqrt{9-3k}}{3}$$

$$x>-2 \Rightarrow x=-1+\frac{\sqrt{9-3k}}{3}$$

(c) **[3]**

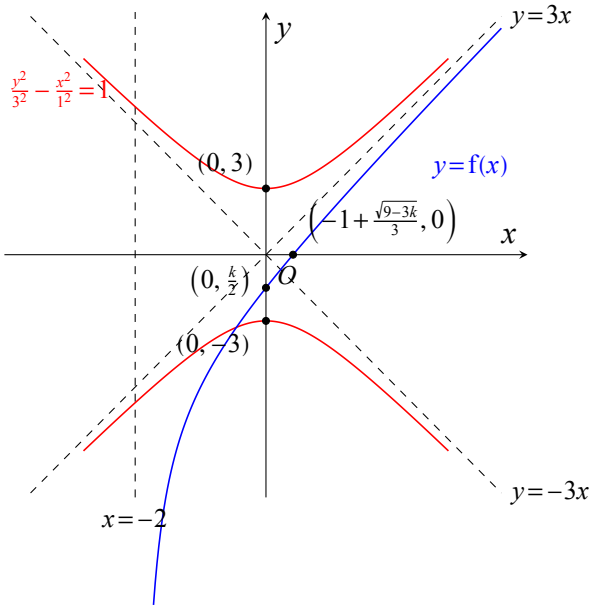
$$(3x^2+6x+k)^2=(x^2+1)\,(9x^2+36x+36) \quad ---(*)$$

$$(3x^2+6x+k)^2=9\,(x^2+1)\,(x+2)^2$$

$$\left(\frac{3x^2+6x+k}{x+2}\right)^2-9x^2=9$$

Draw the hyperbola $\dfrac{y^2}{3^2}-\dfrac{x^2}{1^2}=1$

Asymptotes $y=\pm 3x$, vertices $(0,\pm 3)$



$y=\mathrm{f}(x)$ and $\dfrac{y^2}{3^2}-\dfrac{x^2}{1^2}=1$ have one intersection point

$\therefore$ $(*)$ has one real root

3.4 The curve C is defined by the parametric equations $x=6t-5$ and $y=2t^2+1$, where $t \geqslant \frac{5}{6}$.

(a) Sketch the curve C, labelling the exact coordinates of any axial intercept(s). [1]

(b) Find the equation of the normal to C at the point P where $t=1$. [2]

(c) Find the acute angle between the normal to C at the point P where $t=1$ and the tangent to C at the point M where $t=3$. [2]

(d) The curve C is translated 5 units in the positive x-direction and translated 3 units in the negative y-direction, to form the curve D. Find the equation of D in parametric form. [2]

The curve E is defined by the parametric equations $x=7u$ and $y=\frac{9}{u}$, where $u \neq 0$.

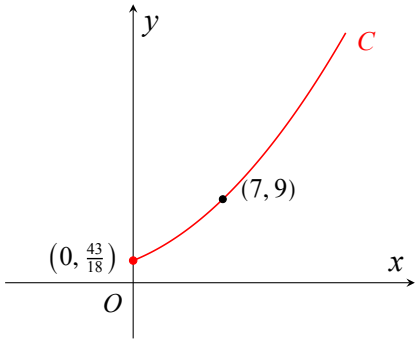
(e) Show that at the point of intersection of the curves C and E, $6t^3-5t^2+3t-34=0$. Deduce that there is only one point of intersection and find the coordinates of this point. [4]

[RVHS 2025 Prelim P1 Q12]

(a) **[1]**

$$t=\frac{5}{6}: \quad x=6\left(\frac{5}{6}\right)-5=0, \quad y=2\left(\frac{5}{6}\right)^2+1=\frac{43}{18}$$

Endpoint (y-intercept): $\left(0, \frac{43}{18}\right)$



(b) **[2]**

$$\frac{dx}{dt}=6, \quad \frac{dy}{dt}=4t$$

$$\frac{dy}{dx}=\frac{4t}{6}=\frac{2}{3}t$$

$t=1$: $\quad \frac{dy}{dx}=\frac{2}{3} \Rightarrow$ gradient of normal $=-\frac{3}{2}$

$t=1$: $\quad x=1,\ y=3$

$$y-3=-\frac{3}{2}(x-1)$$

$$y=-\frac{3}{2}x+\frac{9}{2}$$

(c) **[2]**

$t=3$: $\quad$ gradient of tangent $=\frac{2}{3}(3)=2$

$$\theta_1+\theta_2=\tan^{-1}(2)-\tan^{-1}\left(-\frac{3}{2}\right)=2.089942 \text{ or } 119.7^\circ$$

Acute angle $=\pi-2.089942$ or $180^\circ-119.7^\circ$

$=1.05$ or 60.3°

(d) **[2]**

$$x-5=6t-5 \Rightarrow x=6t$$

$$y+3=2t^2+1 \Rightarrow y=2t^2-2$$

D: $\quad x=6t,\ y=2t^2-2$

(e) **[4]**

$$6t-5=7u \Rightarrow u=\frac{6t-5}{7} \qquad ---(1)$$

$$2t^2+1=\frac{9}{u} \qquad ---(2)$$

Sub (1) into (2): $2t^2+1=\frac{63}{6t-5}$

$$(2t^2+1)(6t-5)=63$$

$$12t^3-10t^2+6t-68=0$$

$$6t^3-5t^2+3t-34=0 \quad \text{(shown)}$$

$$(t-2)(6t^2+7t+17)=0$$

$6t^2+7t+17$: $b^2-4ac=7^2-4(6)(17)=-359<0 \Rightarrow$ no real roots

$\therefore t=2$ only $\Rightarrow$ only one point of intersection

$t=2$: $\quad x=6(2)-5=7,\ y=2(2)^2+1=9$

Point of intersection: $(7, 9)$

3.5 The parametric equations of the curve C are

$$x=\tan\theta-1 \quad \text{and} \quad y=2\sec\theta, \quad \text{where } -\frac{\pi}{2}<\theta<\frac{\pi}{2}.$$

(a) Find $\frac{\mathrm{d}y}{\mathrm{d}x}$ in terms of θ, leaving your answer in a single trigonometric function. [2]

(b) Find the equation of the tangent to C at the point where it cuts the y-axis, leaving your answer in exact form. [3]

(c) Find the Cartesian equation of C. [1]

(d) Find the equations of the asymptotes and state the coordinates of any turning point(s). [3]

(e) The region bounded by the tangent found in part (b), the curve C and the line $x=-1$ is rotated completely about the x-axis. Calculate the volume obtained. [3]

[HCI 2024 Promo Q12]

(a) **[2]**

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\mathrm{d}y}{\mathrm{d}\theta}\div\frac{\mathrm{d}x}{\mathrm{d}\theta}=\frac{2\sec\theta\tan\theta}{\sec^2\theta}=2\sin\theta$$

(b) **[3]**

$$x=0: \quad \tan\theta-1=0 \Rightarrow \tan\theta=1$$

$$\theta=\frac{\pi}{4} \quad \text{since } -\frac{\pi}{2}<\theta<\frac{\pi}{2}$$

$$\theta=\frac{\pi}{4}: \quad y=\frac{2}{\cos\frac{\pi}{4}}=2\sqrt{2}, \quad \frac{\mathrm{d}y}{\mathrm{d}x}=2\sin\frac{\pi}{4}=\sqrt{2}$$

$$\text{Tangent: } y=\sqrt{2}x+2\sqrt{2}$$

(c) **[1]**

$$\tan^2\theta+1=\sec^2\theta$$

$$(x+1)^2+1=\left(\frac{y}{2}\right)^2$$

$$-\frac{\pi}{2}<\theta<\frac{\pi}{2} \Rightarrow 0<\cos\theta\leqslant 1 \Rightarrow \sec\theta\geqslant 1$$

$$\therefore (x+1)^2+1=\left(\frac{y}{2}\right)^2, \quad y\geqslant 2$$

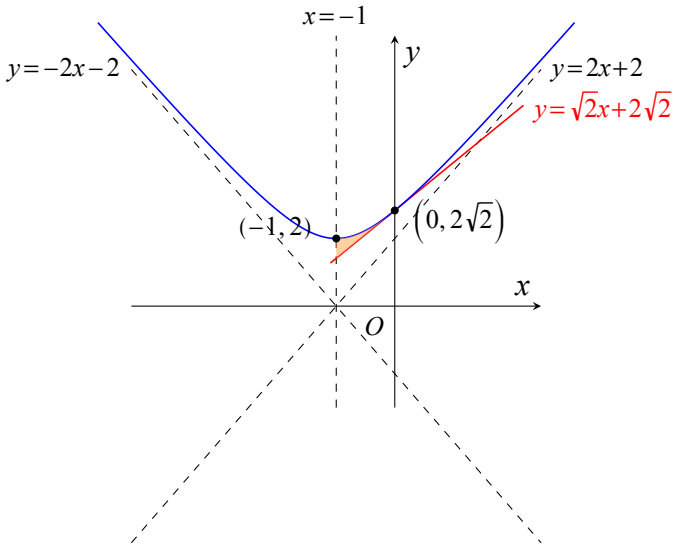
(d) **[3]**

Asymptotes: $\left(\frac{y}{2}\right)^2 = (x+1)^2$

$$\frac{y}{2} = \pm(x+1)$$

$$y = 2x+2 \quad \text{or} \quad y = -2x-2$$

Centre $(-1, 0)$, $y \geqslant 2 \Rightarrow$ turning point $(-1, 2)$



(e) **[3]**

$$\frac{y^2}{4} = (x+1)^2+1$$

$$y^2 = 4(x+1)^2+4$$

$$\text{Volume} = \pi\int_{-1}^{0}\left[4\left((x+1)^2+1\right) - \left(\sqrt{2}x+2\sqrt{2}\right)^2\right]\mathrm{d}x$$

$$= 2.09 \text{ units}^3$$

4 Graphing Transformations

4.1 A curve with equation $y=\mathrm{h}(x)$ undergoes the following sequence of transformations.

A: A translation of 2 units in the positive direction of the x-axis.

B: A scaling parallel to the x-axis by factor of 3.

C: A reflection in the y-axis.

The resultant curve has equation $y=-\dfrac{x^2}{6x+45}$. Find $\mathrm{h}(x)$. [3]

[NJC 2024 Promo Q4(b)]

(b) **[3]**

$$y=-\frac{x^2}{6x+45}$$

C^{-1}: reflection in the y-axis $(x\to -x)$

$$y=-\frac{(-x)^2}{6(-x)+45}=-\frac{x^2}{-6x+45}$$

B^{-1}: scaling parallel to the x-axis by factor $\frac{1}{3}$ $(x\to 3x)$

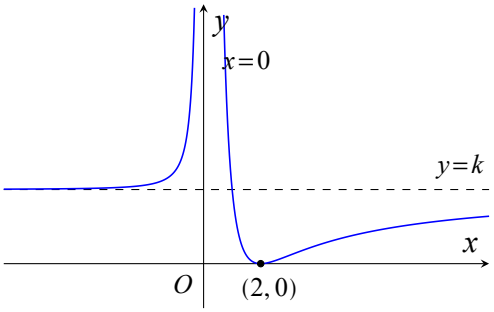
$$y=-\frac{(3x)^2}{-6(3x)+45}=-\frac{9x^2}{-18x+45}=-\frac{x^2}{-2x+5}$$

A^{-1}: translation of 2 units in the negative x-direction $(x\to x+2)$

$$y=-\frac{(x+2)^2}{-2(x+2)+5}=-\frac{(x+2)^2}{-2x+1}$$

$$\mathrm{h}(x)=\frac{(x+2)^2}{2x-1}$$

4.2 The diagram below shows the graph of $y=\mathrm{f}(2x)$. The graph has a turning point at $(2,0)$, and asymptotes with equations $x=0$ and $y=k$.



(a) State a single transformation that will transform the graph of $y=\mathrm{f}(2x)$ onto the graph of $y=\mathrm{f}(2x+4)$. Hence sketch the graph of $y=\mathrm{f}(2x+4)$. [3]

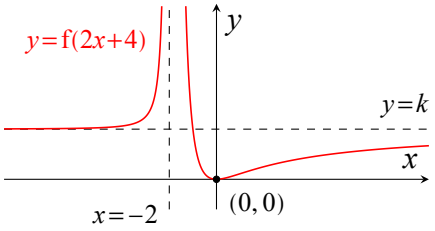
On separate clearly labelled diagrams, sketch the graphs of

(b) $y=\dfrac{1}{\mathrm{f}(2x)}$, [3]

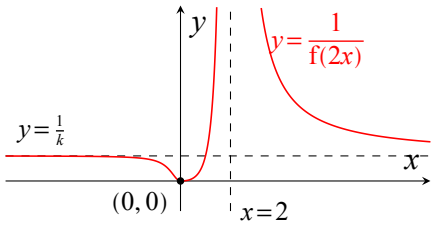
(c) $y=-\mathrm{f}(|x|)$. [3]

[EJC 2025 Prelim P1 Q4]

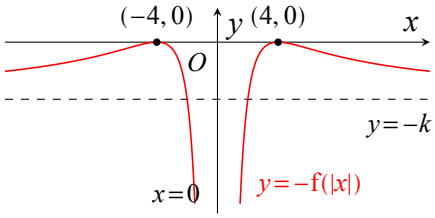
(a) **[3]**



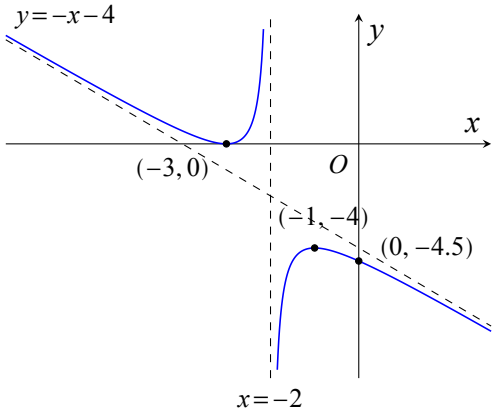
(b) **[3]**



(c) **[3]**



4.3



The diagram shows the curve $y = \mathrm{f}(x)$. The curve crosses the x-axis at its minimum point $(-3, 0)$. It has a maximum turning point $(-1, -4)$ and crosses the y-axis at the point $(0, -4.5)$. The asymptotes of the curve are $x = -2$ and $y = -x - 4$.

On separate diagrams, sketch the graphs of

(a) $y = \mathrm{f}'(x)$, [3]

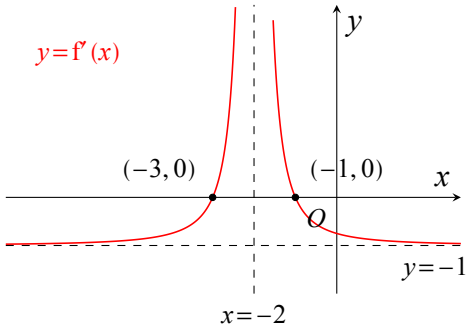
(b) $y = \dfrac{1}{\mathrm{f}(x)}$, [3]

(c) $y = |\mathrm{f}(x)|$, [3]

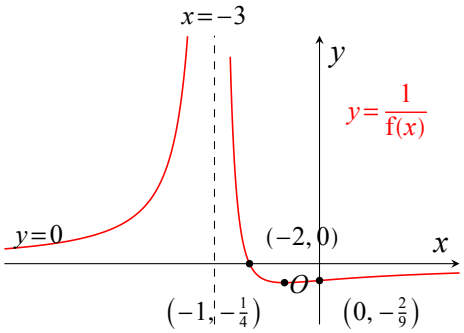
indicating the equations of any asymptotes, turning points and intersections with the axes.

[CJC 2025 Promo Q8]

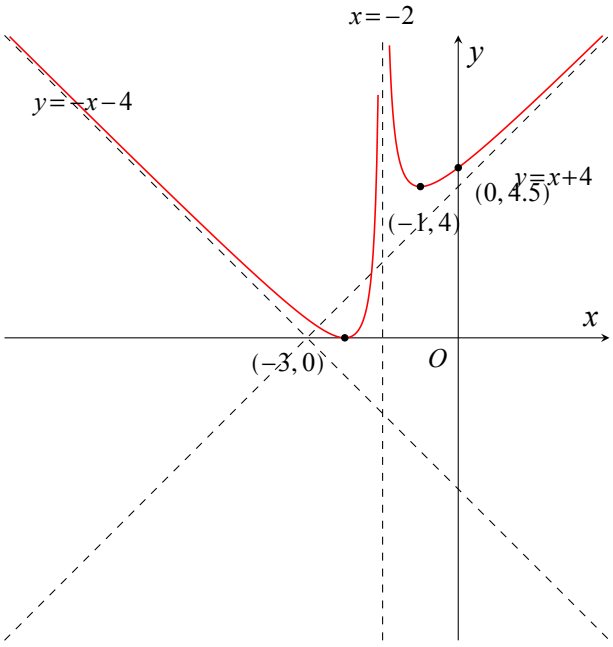
(a)


y = f′(x)
y
x
(−3, 0)
(−1, 0)
O
y = −1
x = −2

(b)


x = −3
y
y = 1/f(x)
y = 0
(−2, 0)
x
O
(−1, −1/4)
(0, −2/9)

(c)


x = −2
y
y = −x − 4
y = x + 4
(0, 4.5)
(−1, 4)
x
(−3, 0)
O

4.4 Express $\frac{9x+5}{3x+2}$ in the form $a-\frac{b}{3x+2}$ where a and b are positive constants.

Hence, describe a sequence of transformations which transform the graph of $y=\frac{9x+5}{3x+2}$ to the graph of $y=\frac{1}{x}$. [5]

[RVHS 2026 Prelim P1 Q10(b)]

(b) **[5]**

$$\frac{9x+5}{3x+2}=3-\frac{1}{3x+2} \qquad (a=3,\ b=1)$$

$$y=3-\frac{1}{3x+2} \xrightarrow{y\to y+3} y=-\frac{1}{3x+2}$$
$$\xrightarrow{y\to -y} y=\frac{1}{3x+2}$$
$$\xrightarrow{x\to \frac{x}{3}} y=\frac{1}{x+2}$$
$$\xrightarrow{x\to x-2} y=\frac{1}{x}$$

1. Translate 3 units in the negative y-direction
2. Reflect about the x-axis
3. Scale by factor 3 parallel to the x-axis
4. Translate 2 units in the positive x-direction

5 Functions

5.1 The function f is defined by

$$\mathrm{f}: x \to \left|\frac{2x+4}{3-x}\right|, \qquad x \in \mathbb{R}, \qquad x \neq 3.$$

(a) Sketch the graph of $y = \mathrm{f}(x)$, giving the equations of any asymptotes and the coordinates of the points where the curve meets the axes. [3]

(b) Hence state the range of f. [1]

(c) Explain why the function f^2 does not exist. [1]

The domain of f is further restricted to $x \leqslant a$, where a is a constant.

(d) State the greatest value of a such that the function f^{-1} exists. [1]

(e) Hence find $\mathrm{f}^{-1}(x)$ and state its domain. [4]

[2023 A-Level P1 Q7]

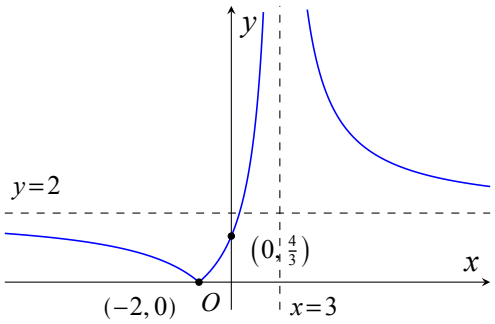
(a) **[3]**

$$y = \left|\frac{2x+4}{3-x}\right| = \left|-2+\frac{10}{3-x}\right|$$

Asymptotes: $y = |-2| = 2, \quad x = 3$

$$x = 0: \quad y = \left|\frac{4}{3}\right| = \frac{4}{3}$$

$$y = 0: \quad x = -2$$



(b) **[1]**

$$\mathrm{R_f} = [0, \infty)$$

(c) **[1]**

$$R_f = [0, \infty), \quad D_f = (-\infty, 3) \cup (3, \infty)$$

$$3 \in R_f,\ 3 \notin D_f \Rightarrow R_f \not\subseteq D_f$$

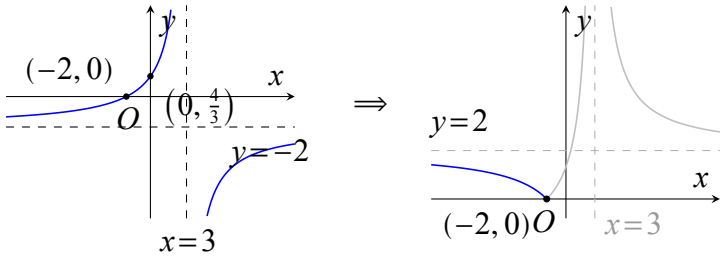
$\therefore f^2$ does not exist

(d) **[1]**

$$a = -2$$

(e) **[4]**

$$x \leqslant -2: \quad f(x) = \left|\frac{2x+4}{3-x}\right| = -\frac{2x+4}{3-x}$$



$$y = f(x)$$

$$-3y + xy = 2x + 4$$

$$x(y-2) = 4 + 3y$$

$$x = \frac{4+3y}{y-2}$$

$$f^{-1}(x) = \frac{4+3x}{x-2}, \qquad D_{f^{-1}} = [0, 2)$$

5.2 The function f is defined by

$$\mathrm{f}: x \mapsto \frac{ax-3}{5x-4}, \text{ for } x \in \mathbb{R},\ x \neq \frac{4}{5}, \text{ and } a \in \mathbb{R}.$$

(a) Find the range of f in terms of a. [1]

The function f is such that $\mathrm{f}(x) = \mathrm{f}^{-1}(x)$ for all x in the domain of f.

(b) Find the value of a. [2]

(c) Hence find $\mathrm{f}^{2025}(2)$. [2]

Another function g is defined by

$$\mathrm{g}: x \mapsto \left(x-\frac{1}{5}\right)^2 + \frac{2}{5}, \text{ for } x \in \mathbb{R},\ 0 < x < \frac{3}{5}.$$

(d) Find the range of fg. [2]

[HCI 2025 Prelim P1 Q8]

(a) **[1]**

Horizontal asymptote: $y = \frac{a}{5}$

$$\mathrm{R_f} = \mathbb{R} \setminus \left\{\frac{a}{5}\right\}$$

(b) **[2]**

$$y = \frac{ax-3}{5x-4}$$
$$5xy - 4y = ax - 3$$
$$5xy - ax = 4y - 3$$
$$x = \frac{4y-3}{5y-a}$$
$$\mathrm{f}^{-1}(x) = \frac{4x-3}{5x-a}, \quad x \in \mathbb{R},\ x \neq \frac{a}{5}$$
$$\mathrm{f}(x) = \mathrm{f}^{-1}(x) \Rightarrow a = 4$$

(c) **[2]**

$$\begin{aligned} f^{-1}(x) = f(x) \;\Rightarrow\; f^2(x) &= ff^{-1}(x) = x \\ f^{2025}(2) &= f^2\left(f^{2023}(2)\right) \\ &= f^2\left(f^{2021}(2)\right) \\ &= \cdots \\ &= f(2) \\ &= \frac{4(2)-3}{5(2)-4} = \frac{5}{6} \end{aligned}$$

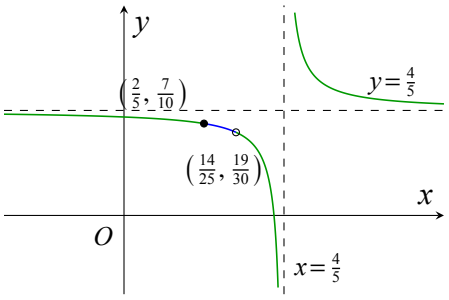
(d) **[2]**

$$R_g = \left[\frac{2}{5}, \frac{14}{25}\right)$$

$$f(x) = \frac{4x-3}{5x-4}$$

$$\left(0, \frac{3}{5}\right) \xrightarrow{g} \left[\frac{2}{5}, \frac{14}{25}\right) \xrightarrow{f} \left(\frac{19}{30}, \frac{7}{10}\right]$$

$$R_{fg} = \left(\frac{19}{30}, \frac{7}{10}\right]$$



5.3 The function f is given by

$$\mathrm{f}(x)=\begin{cases}2+\sqrt{2^2-(x-2)^2}, & 0<x\leqslant 4,\\ 2-\sqrt{2^2-(x-6)^2}, & 4<x\leqslant 8.\end{cases}$$

It is given that $\mathrm{f}(x)=\mathrm{f}(x+8)$ for all real values of x.

(a) On the diagram in the Printed Answer Book, sketch the graph of $y=\mathrm{f}(x)$ for $-6\leqslant x\leqslant 7$, indicating clearly the coordinates of the end points and the points where the graph cuts the axes. [3]

(b) Without integrating, write down the exact area of the region bounded by $y=\mathrm{f}(x)$, the line $x=4$, the x-axis and the y-axis. [1]

[VJC 2025 Prelim P1 Q5(a), (b)]

(a) **[3]**

$y=2+\sqrt{2^2-(x-2)^2}$

$y-2=\sqrt{2^2-(x-2)^2}$

$(x-2)^2+(y-2)^2=2^2$

Centre $(2,2)$, radius 2.

Since it is a positive square root, it is the top half of this circle.

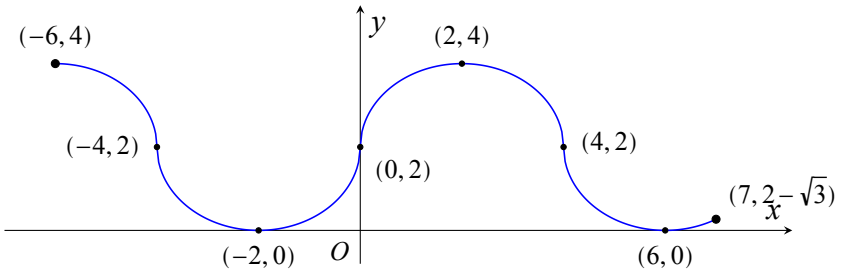
$y=2-\sqrt{2^2-(x-6)^2}$

$y-2=-\sqrt{2^2-(x-6)^2}$

$(x-6)^2+(y-2)^2=2^2$

Centre $(6,2)$, radius 2.

Since it is a negative square root, it is the bottom half of this circle.



(b) **[1]**

$$\text{Area}=2\times 4+\frac{1}{2}\pi(2^2)=8+2\pi$$

5.4 The function f is defined by

$$\mathrm{f}(x) = 1 - ax,\ x \in \mathbb{R},\ \text{where } a \text{ is a real constant.}$$

The function g is defined by

$$\mathrm{g}(x) = \begin{cases} x^2 + 3 & \text{for} \quad 0 \leqslant x \leqslant 2, \\ 7 - x & \text{for} \quad 2 < x \leqslant 7. \end{cases}$$

(i) Find the set of possible values of a such that f^{-1} exist. [1]

(ii) If $a = 2$, describe a sequence of transformations that transform the graph of $y = \mathrm{g}(x)$ onto the graph of $y = \mathrm{fg}(x)$. [3]

(iii) The function gg is defined by

$$\mathrm{gg}(x) = \begin{cases} \mathrm{h}_1(x) & \text{for} \quad 0 \leqslant x \leqslant 2, \\ \mathrm{h}_2(x) & \text{for} \quad 2 < x < 5, \\ \mathrm{h}_3(x) & \text{for} \quad 5 \leqslant x \leqslant 7. \end{cases}$$

Find $\mathrm{h}_1(x)$, $\mathrm{h}_2(x)$ and $\mathrm{h}_3(x)$. [3]

[ACJC 2022 Prelim P2 Q1]

(i) **[1]**

$$a \in \mathbb{R} \setminus \{0\}$$

(ii) **[3]**

$$y = \mathrm{fg}(x) = 1 - 2\mathrm{g}(x)$$

Reflection of the graph of $y = \mathrm{g}(x)$ in the x-axis, followed by scaling by scale factor 2 parallel to the y-axis, followed by translation 1 unit in the positive y-direction.

OR

Translation of the graph of $y = \mathrm{g}(x)$ by $\frac{1}{2}$ unit in the negative y-direction followed by reflection in the x-axis followed by scaling by scale factor 2 parallel to the y-axis.

OR

Translation of the graph of $y = \mathrm{g}(x)$ by $\frac{1}{2}$ unit in the negative y-direction followed by scaling by scale factor 2 parallel to the y-axis followed by reflection in the x-axis.

(iii) **[3]**

$$\mathrm{gg}(x)=\begin{cases}7-(x^2+3) & \text{for} \quad 0\leqslant x\leqslant 2\\ 7-(7-x) & \text{for} \quad 2<x<5\\ (7-x)^2+3 & \text{for} \quad 5\leqslant x\leqslant 7\end{cases}$$

$$\therefore \mathrm{gg}(x)=\begin{cases}4-x^2 & \text{for} \quad 0\leqslant x\leqslant 2\\ x & \text{for} \quad 2<x<5\\ x^2-14x+52 & \text{for} \quad 5\leqslant x\leqslant 7\end{cases}$$

5.5 Functions f and g are defined by

$$\mathrm{f}: x \to \frac{4}{(x-4)^2}, \quad x \in \mathbb{R},\ x \neq 4,$$

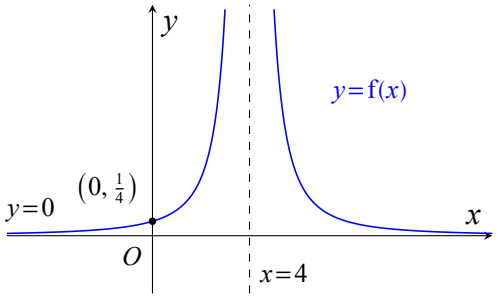
$$\mathrm{g}: x \to \ln\left(1+\frac{1}{x}\right), \quad x \in \mathbb{R},\ x > 0.$$

(a) Sketch the graph of $y = \mathrm{f}(x)$, stating the equations of any asymptotes, the coordinates of the points where it crosses the axes and the coordinates of the turning points, if any. [2]

(b) Show that gf exists. Hence find the rule, domain and range of gf. [4]

[CJC 2025 Prelim P2 Q2(a), (b)]

(a) **[2]**



(b) **[4]**

For gf to exist, $\mathrm{R_f} \subseteq \mathrm{D_g}$,

$$\mathrm{R_f} = (0, \infty)$$

$$\mathrm{D_g} = (0, \infty)$$

Since $\mathrm{R_f} \subseteq \mathrm{D_g}$, gf exists (shown)

$$\begin{aligned}\mathrm{gf}(x) &= \mathrm{g}\left[\frac{4}{(x-4)^2}\right] \\ &= \ln\left[1+\left(\frac{1}{\frac{4}{(x-4)^2}}\right)\right] \\ &= \ln\left[1+\frac{(x-4)^2}{4}\right]\end{aligned}$$

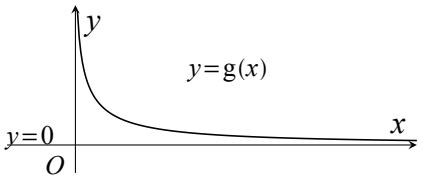
$$\mathrm{D_{gf}} = \mathrm{D_f} = (-\infty, 4) \cup (4, \infty)$$

OR

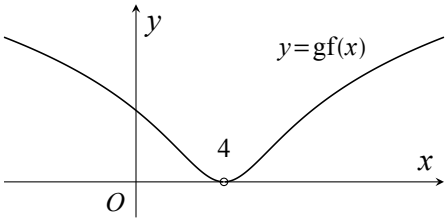
$$\mathrm{D_{gf}} = \mathrm{D_f} = \mathbb{R} \setminus \{4\}$$

Method 1: Mapping

$$\mathrm{D_f} = (-\infty, 4) \cup (4, \infty) \xrightarrow{\mathrm{f}} \mathrm{R_f} = (0, \infty) \xrightarrow{\mathrm{g}} \mathrm{R_{gf}} = (0, \infty)$$



Method 2: Sketch $y=\mathrm{gf}(x)$ for given domain



$\mathrm{R}_{\mathrm{gf}}=(0,\infty)$

5.6 The function g is defined by $g(0)=g(1)=1$, $g(2n)=2+g(n)$ and $g(2n+1)=1+g(2n)$, for integer n.

(i) Find $g(4)$, $g(7)$ and $g(12)$.

(ii) Does g have an inverse? Justify your answer. [3]

[2016 A-Level P1 Q10(b)]

(i)

$$\begin{aligned} g(4) &= 2+g(2) \\ &= 2+2+g(1) \\ &= 2+2+1+g(0) \\ &= 2+2+1+1 \\ &= 6 \end{aligned}$$

$$\begin{aligned} g(7) &= 1+g(6) \\ &= 1+2+g(3) \\ &= 1+2+1+g(2) \\ &= 1+2+1+2+1+1 \\ &= 8 \end{aligned}$$

$$\begin{aligned} g(12) &= 2+g(6) \\ &= 2+2+1+2+1+1 \\ &= 9 \end{aligned}$$

(ii) **[3]**

$$\begin{aligned} g(8) &= 2+g(4) \\ &= 2+6 \\ &= 8 \\ &= g(7) \end{aligned}$$

Since $g(8)=g(7)=8$, g is not one-one, so g^{-1} does not exist.